c) That mathematics is not easy or logic is difficult.

Soln: Conclusion: $\neg r \vee p$

Steps	Reason
1. $p \vee \neg q$	Premise
2. $r \to \neg p$	Premise
3. $\neg r \vee \neg p$	Conditional-disjunction equivalence on 2

Clause C1: $p \vee \neg q$ C2: $\neg r \vee \neg p$ C3:

$\neg(\neg r \vee p) \equiv \neg(\neg r) \wedge \neg p$ (De Morgan's Law) $\equiv r \wedge \neg p$ (Double-negation Law)

C3: r C4: $\neg p$

By Resolution on C3, C2 we get C5: $\neg p$

By Resolution on C4, C5 we get C6: $\neg p$

By Resolution on C6, C1 we get C7: $\neg q$

We cannot reach a contradiction from C7 and other clauses

$\therefore$ The conclusion is not valid.

d) That logic is not difficult or mathematics is not easy.

Soln: Conclusion: $\neg p \vee \neg r$

Steps	Reason
1. $p \vee \neg q$	Premise
2. $r \to \neg p$	Premise
3. $\neg r \vee \neg p$	Conditional-disjunction equivalence on 2
4. $\neg p \vee \neg r$	Commutative law on 3

e) That if not many students like logic, then either mathematics is not easy or logic is not difficult.

Soln: Conclusion: $\neg q \to (\neg r \vee \neg p)$